

**KENYATTA UNIVERSITY DEPARTMENT OF GEOGRAPHY AGE201/211:
PHYSICAL GEOGRAPHY II**

SECTION A: SHORT QUESTIONS

Question 1

(a) Differentiate between Coalescence model and Bergeron-Findeise process (3 marks)

- **Coalescence Model:** This process occurs in warm clouds where larger water droplets collide and merge with smaller droplets, growing in size until they fall as raindrops.
- **Bergeron-Findeise Process:** This process occurs in cold clouds where ice crystals grow at the expense of surrounding super cooled water droplets, eventually becoming large enough to fall as snow or rain.

(b) Suggest three ways in which flood frequency analyses may be useful (3 marks)

1. **Risk Assessment:** Helps in evaluating the likelihood and severity of future floods, aiding in disaster preparedness.
2. **Infrastructure Planning:** Informs the design and construction of flood-resistant structures and drainage systems.
3. **Insurance and Policy Making:** Assists in setting insurance premiums and formulating policies for flood-prone areas.

(c) (i) What is a river gauging station (RGS)? (3 marks)

- **River Gauging Station:** A site on a river where measurements of water discharge, level, and flow velocity are regularly recorded.

(ii) Which is the best location for RGS? (3 marks)

- The best location for a River Gauging Station is a stable section of the river with minimal turbulence and human interference, where the riverbed and banks are consistent and not prone to significant changes.

(d) Differentiate between connate and juvenile water (3 marks)

- **Connate Water:** Water that has been trapped in the pores of a sedimentary rock since its formation.
- **Juvenile Water:** Water that originates from deep within the Earth's mantle, usually associated with volcanic activity and emerging to the surface.

(e) Calculate the porosity of a 250 g rock that contains 65 g of water when 55% of the pores are full of water (3 marks) - Porosity Calculation:

1. Weight of water in pores = 65 g.
2. Porosity (ϕ) = (Volume of pores / Total volume of rock) $\times$ 100.
3. Since 55% of the pores are full, the total pore volume = 65 g / 0.55.
4. Total volume of water = 118.18 g (since 1 g water = 1 cm³, the volume is 118.18 cm³).
5. Porosity (ϕ) = (118.18 cm³ / 250 cm³) $\times$ 100 = 47.27%.

(f) Explain the types of soil moisture in relation to their availability for use by plants (3 marks)

- **Gravitational Water:** Water that drains through the soil due to gravity and is usually not available to plants.
- **Capillary Water:** Water held in the soil pores against gravity, available for plant uptake.
- **Hygroscopic Water:** Water that forms a thin film around soil particles, not available to plants.

(g) Highlight three theories on why Rift valley lakes are rising in sea levels (3 marks)

1. **Increased Rainfall:** Higher precipitation rates due to climate change leading to more water inflow.
2. **Tectonic Activity:** Movements in the Earth's crust affecting the lake's basin and water levels.
3. **Deforestation and Land Use Changes:** Reduced vegetation cover leading to higher runoff and decreased water absorption by the soil.

(h) Explain how a glacier is formed (3 marks)

- A glacier forms through the accumulation, compaction, and recrystallization of snow over many years. As snow accumulates, it compresses and transforms into firn and eventually into dense glacial ice, which flows due to gravity.

(i) Highlight three hazards associated with slope processes (3 marks)

1. **Landslides:**
3. **Mudslides:** Flow of water- soil down a slope, causing significant damage.

(j) State three factors that make wind an effective agent of desert erosion (3 marks)

1. **Lack of Vegetation:** Sparse vegetation in deserts allows wind to easily pick up and transport sand and soil particles.
2. **Dry Conditions:** Dry, loose particles are more susceptible to wind erosion.
3. **Frequent Windstorms:** Strong and persistent winds common in desert areas enhance erosion processes.

SECTION B: HYDROSPHERE Question 2

(a) A soil sample has a weight of 0.7 kg and the volume was found to be $3.5 \times 10^{-4} \text{ m}^3$. After drying out the weight was reduced to 0.6 kg. The particle specific gravity test gave 2.6.

Determine the following:

(i) Moisture Content:

(ii) Dry Density

(iii) Void Ratio:

(iv) Bulk Density:

(vi) Degree of Saturation:

- Degree of Saturation (S) can be calculated using moisture content and porosity.

(b) State FOUR challenges of rainwater harvesting in your home county (8 marks)

1. **Contamination Risks:** Possibility of pollutants entering the water during collection and storage.
2. **High Initial Costs:** Installation of rainwater harvesting systems can be expensive.

3. **Limited Storage Capacity:** Inadequate storage facilities can limit the amount of water that can be harvested.
4. **Seasonal Variability:** Inconsistent rainfall patterns may affect the reliability of harvested water.

SECTION C: LITHOSPHERE Question 4 (a) Explain FIVE factors that can interrupt Morris Davis cycle of erosion (10 marks)

1. **Tectonic Activity:** Uplift and subsidence can alter the landscape, affecting the erosion cycle.
2. **Climate Change:** Changes in climate can influence weathering, erosion, and sediment transport processes.
3. **Human Activities:** Urbanization, deforestation, and mining can disrupt natural erosion cycles.
4. **Vegetation Cover:** Changes in vegetation can stabilize or destabilize slopes, affecting erosion rates.
5. **Sea Level Changes:** Rising or falling sea levels can alter the base level of erosion.

(b) Using diagrams, describe FIVE features formed by wave erosion at the Indian Ocean (10 marks)

1. **Sea Cliffs:** Steep rock faces formed by the erosive action of waves.
2. **Wave-Cut Platforms:** Flat areas at the base of cliffs created by wave erosion.
3. **Sea Arches:** Natural arches formed when waves erode through a headland.
4. **Sea Stacks:** Isolated pillars of rock formed from the collapse of sea arches.
5. **Blowholes:** Holes formed in cliffs where waves force water and air through cracks.

Question 5 (a) Describe FIVE factors affecting slope stability (10 marks)

1. **Slope Angle:** Steeper slopes are more prone to instability.
2. **Soil and Rock Composition:** Certain soil and rock types are more prone to erosion and landslides.

3. **Water Content:** High water content can reduce soil cohesion and increase the risk of landslides.
4. **Vegetation Cover:** Vegetation can stabilize slopes by holding soil in place with roots.
5. **Human Activities:** Construction, deforestation, and mining can destabilize slopes.

(b) Using FIVE examples, explain how weathering has affected the economic activities in the county of your choice (10 marks)

1. **Agriculture:** Weathering breaks down rocks into fertile soil, supporting agriculture.
2. **Mining:** Weathering processes can concentrate valuable minerals, making mining feasible.
3. **Tourism:** Weathered landscapes, such as gorges and canyons, can attract tourists.
4. **Construction:** Weathering affects the durability and stability of construction materials.
5. **Water**